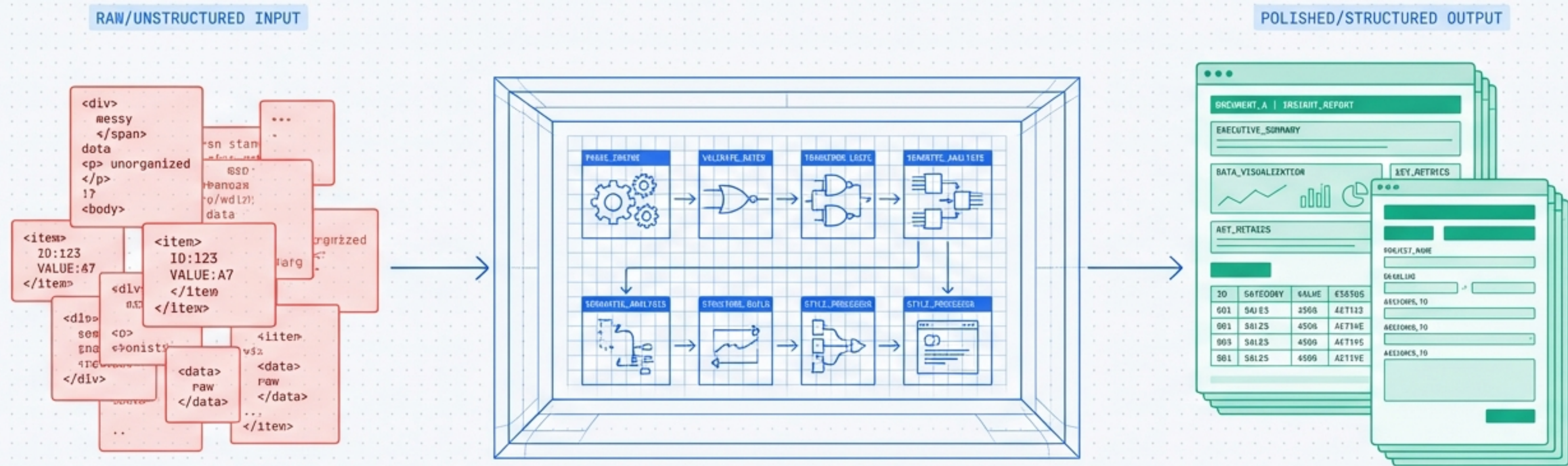


The HTML Transformation Workbench

From Raw Data to Structured Insight



Bringing Control and Analysis to Web Content

Core Concept

A web-based workbench for loading, transforming, and saving HTML pages. Users apply **content filters**, **style modifications**, and **LLM-based analysis** with real-time before/after verification.

Five Pillars



Content Moderation

Remove negative content, blur images, filter by topic relevance.



Style Manipulation

Modify CSS, restructure layouts, strip inline styles.



LLM Processing

Deep content analysis, summarisation, and translation via AI.



Visual Debugging

Real-time previews of transformations before committing changes.

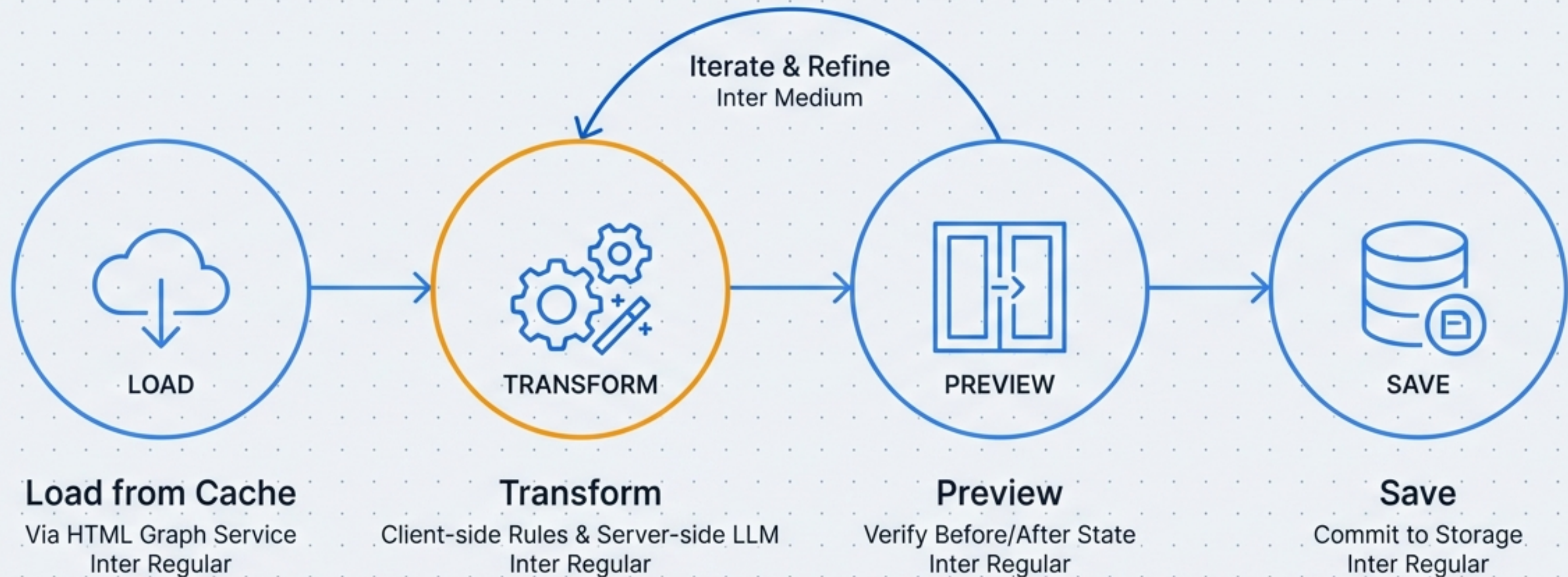


Workflow Automation

Chain multiple transformations into reusable profiles.

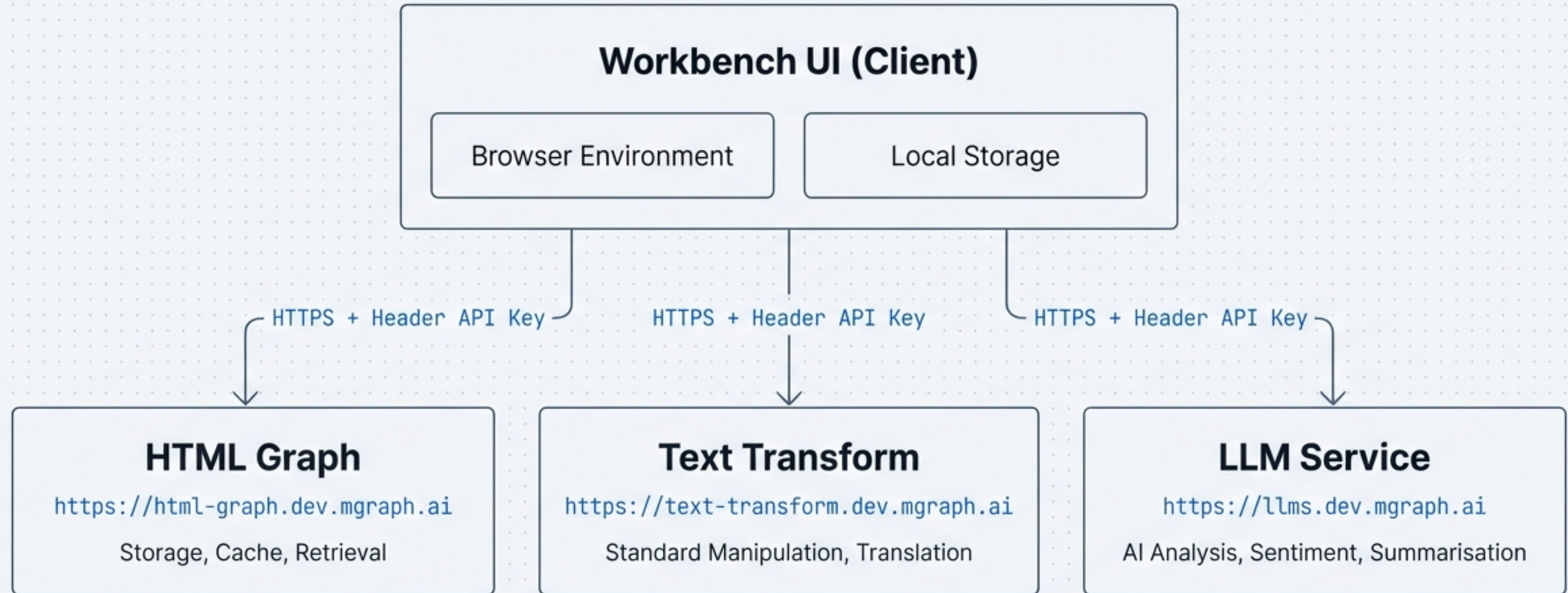
The Transformation Cycle

Iterative Non-Destructive Workflow



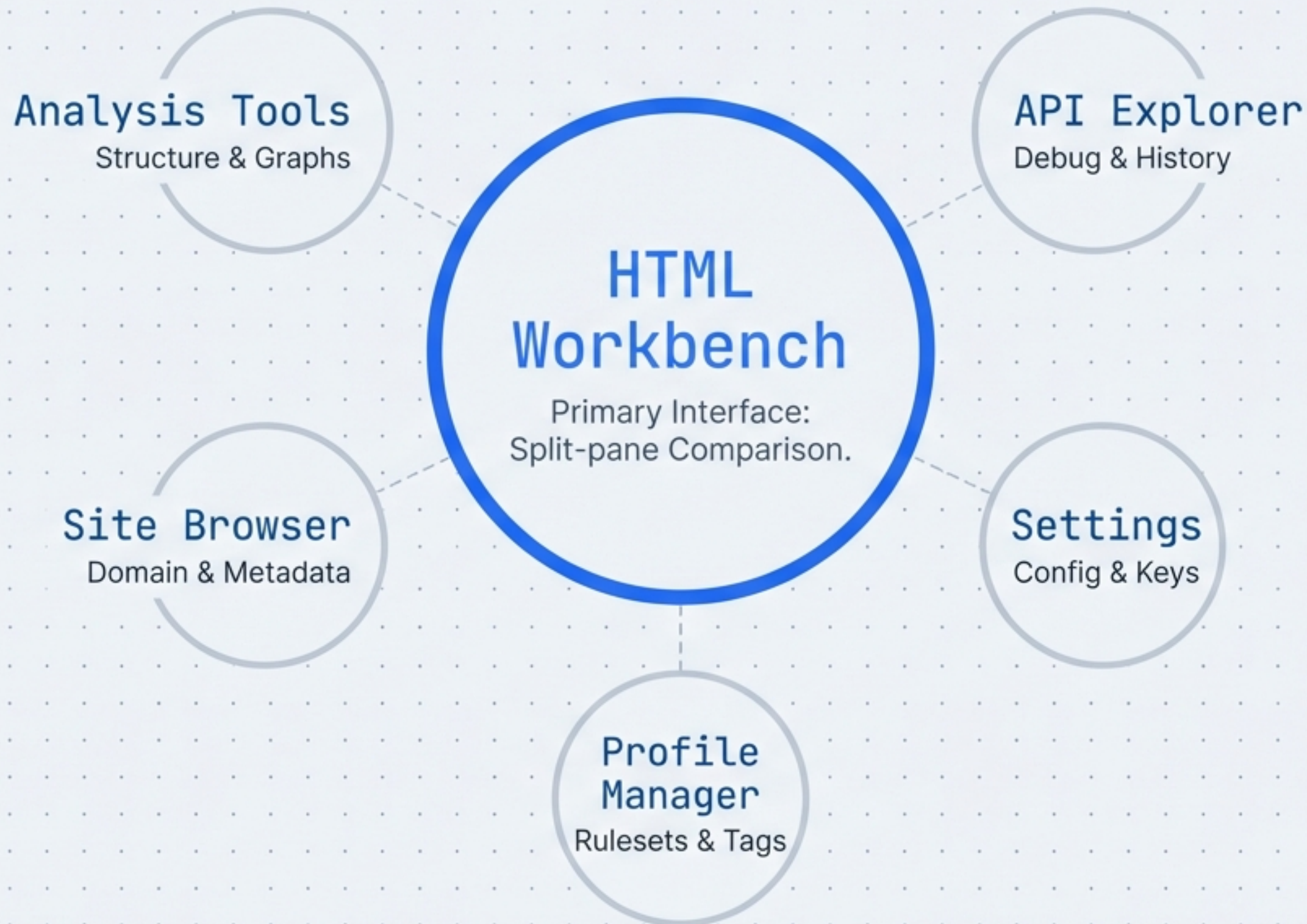
The cycle is designed for iteration. Users can test transformations non-destructively in the 'Preview' stage before saving to the permanent cache.

System Architecture and Service Integration



Event-Driven Architecture: Global events trigger updates across components upon state changes.

The Mini-App Constellation



The Mechanics of Transformation

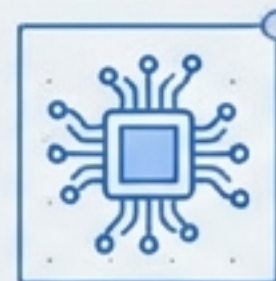
Client-Side (JavaScript)



Speed, Structure, & Style

- Remove Elements (CSS Selectors)
- Blur Images
- Simplify Styles (Strip Inline CSS)
- Remove Ads
- Extract Main Content

Server-Side (LLM & Services)



Intelligence, Meaning, & Context

- Sentiment Filter (Remove Negative)
- Topic Filter (Relevance)
- Summarise (Abstract Generation)
- Translate (Language)

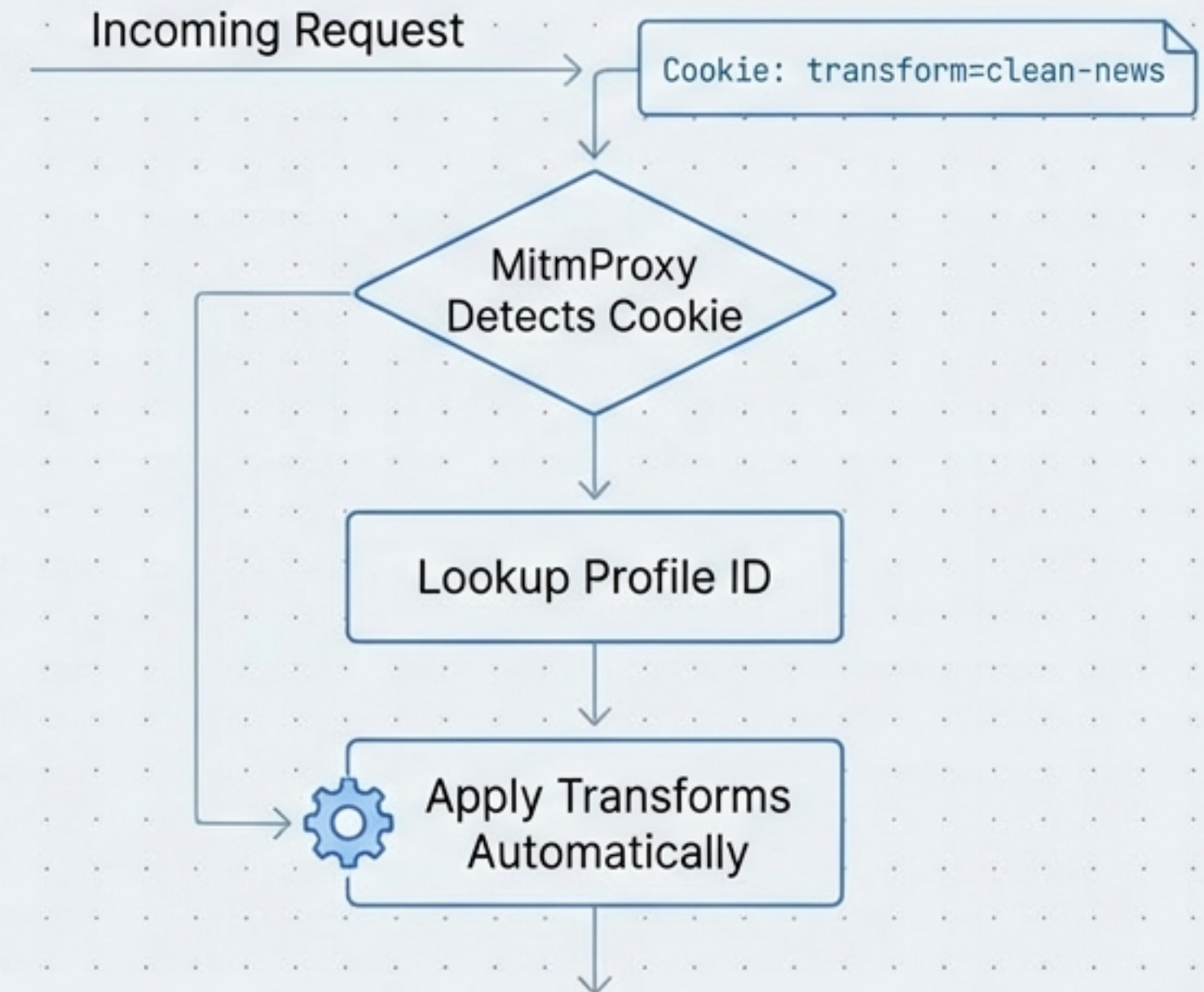
— Integration: Server-side transforms leverage ``html-sentiment-filter`` and ``html-topic-filter`` FLeTs to process content and return new cache IDs.

Automation via Transformation Profiles



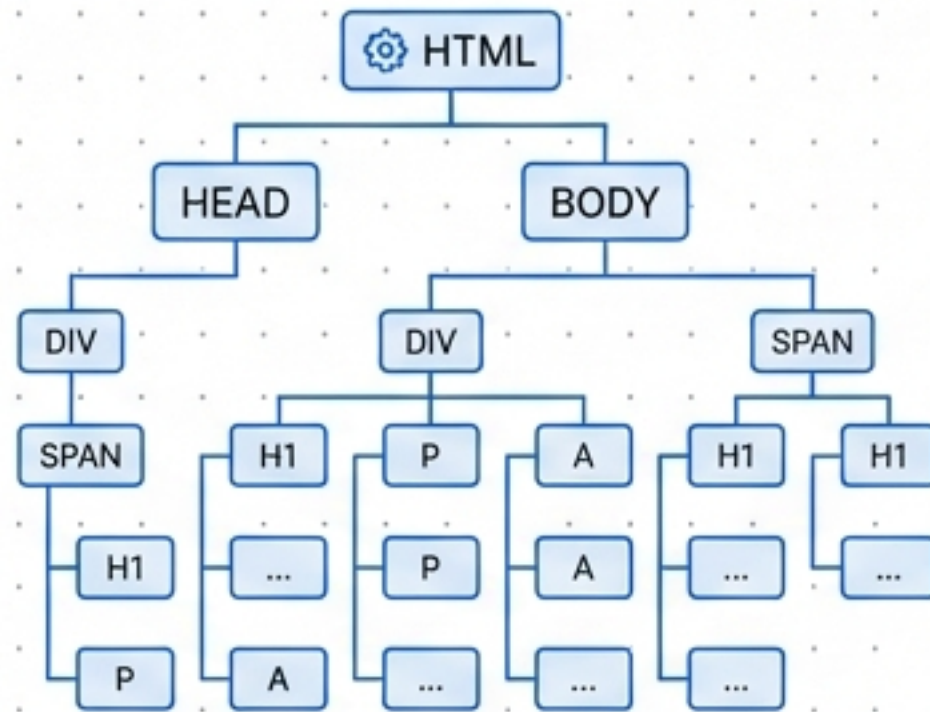
A Profile is a saved, reusable configuration defining which transformations to apply and their parameters. Users can tag, search, and batch-apply these profiles.

MitmProxy Integration Workflow



Deep Page Analysis

Structure



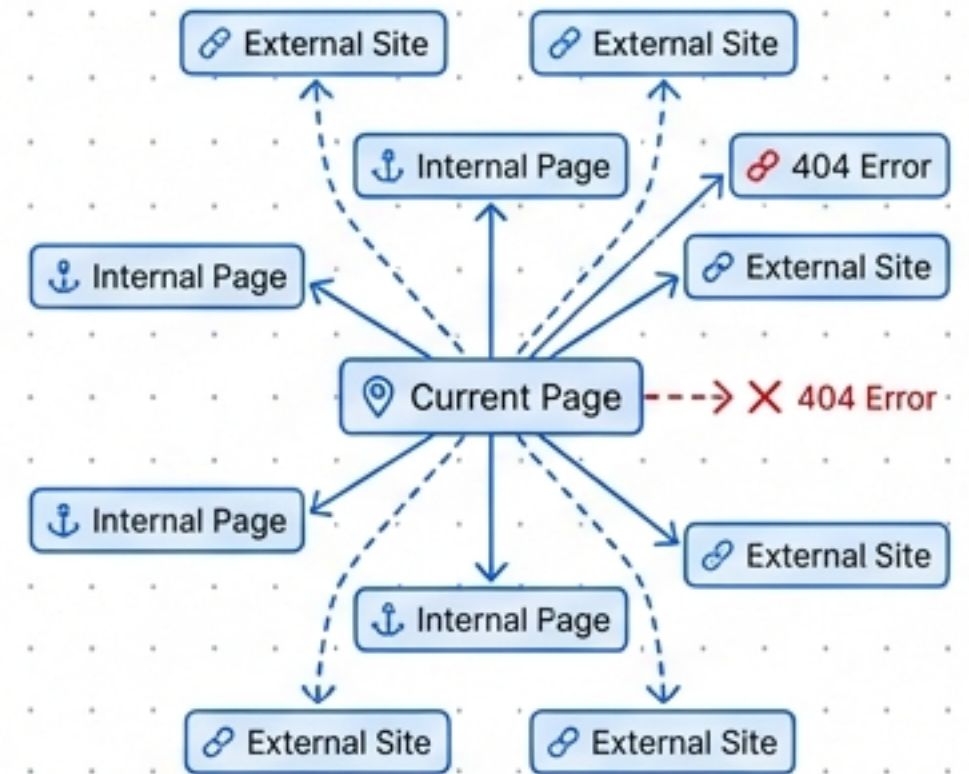
DOM Tree Visualization, Element Counts, Hierarchy Mapping.

Content (LLM)



Word Count, Key Topic Extraction, Sentiment Scoring, Readability Index.
Inter Medium Slate Grey

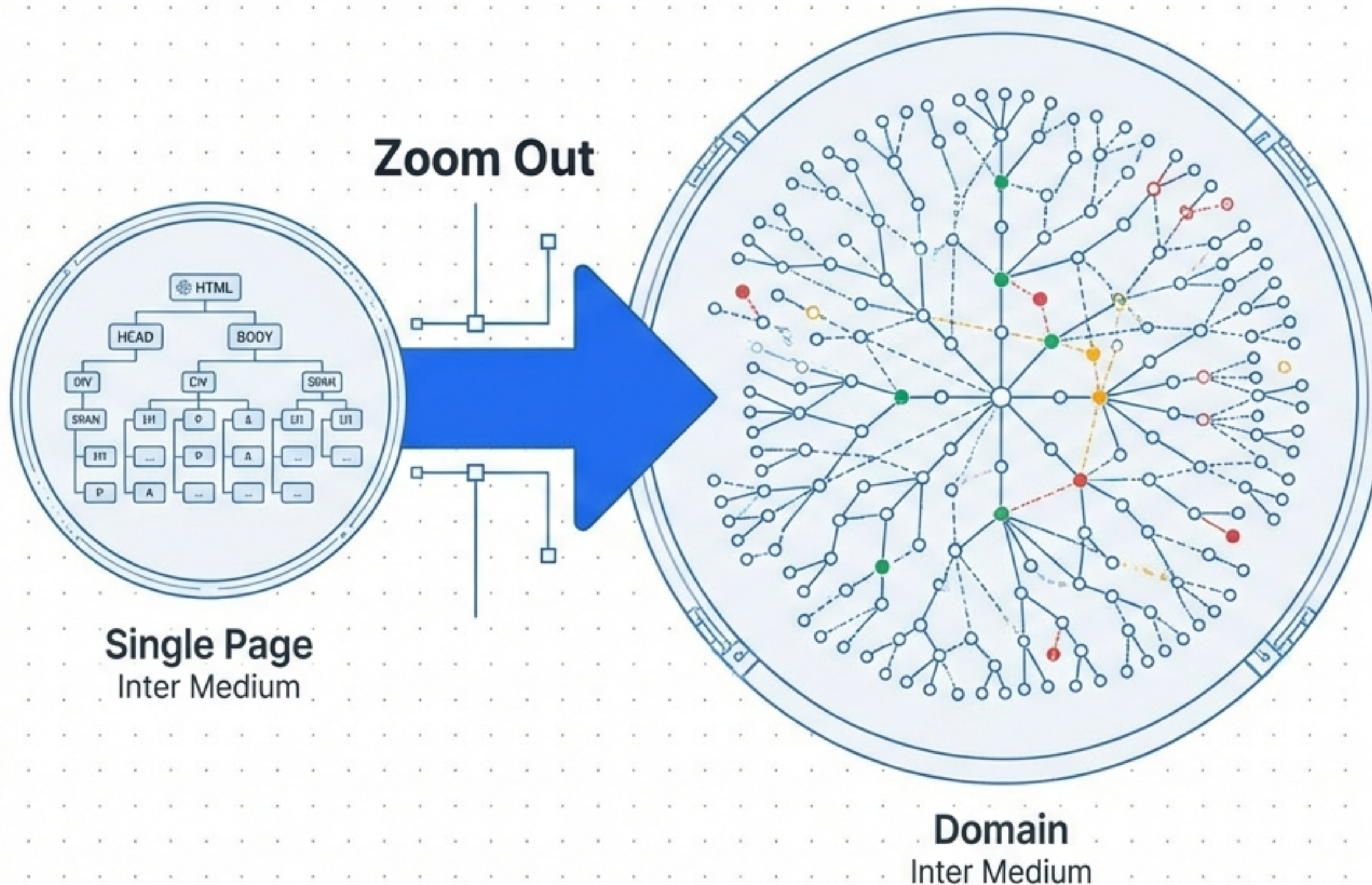
Links



Internal vs. External Mapping, Broken Path Identification.

Analysis results are saved as distinct data files on the entity for historical comparison.

Site-Wide Operations and Intelligence



- **Site Map Generation:** Visualizing complete page hierarchy.
- **Content Overview:** Aggregated topic distribution.
- **Link Graph:** Visualising internal linking structures.
- **Site Comparison:** Tracking changes over time (Diff).
- **Batch Processing:** Apply Transformation Profiles to entire domains.

Query Pattern: `{domain}/{path}` used to access all pages for a specific site.
JetBrains Mono Regular

Roadmap Phase 1 — Foundation

Establishing the Framework



Shell Component: Global nav & routing (``workbench-shell.js``)



Settings Panel: API Config (``settings-panel.js``)



Simple HTML Test: Input Cache
Key -> Load -> Save

SUCCESS CRITERIA:



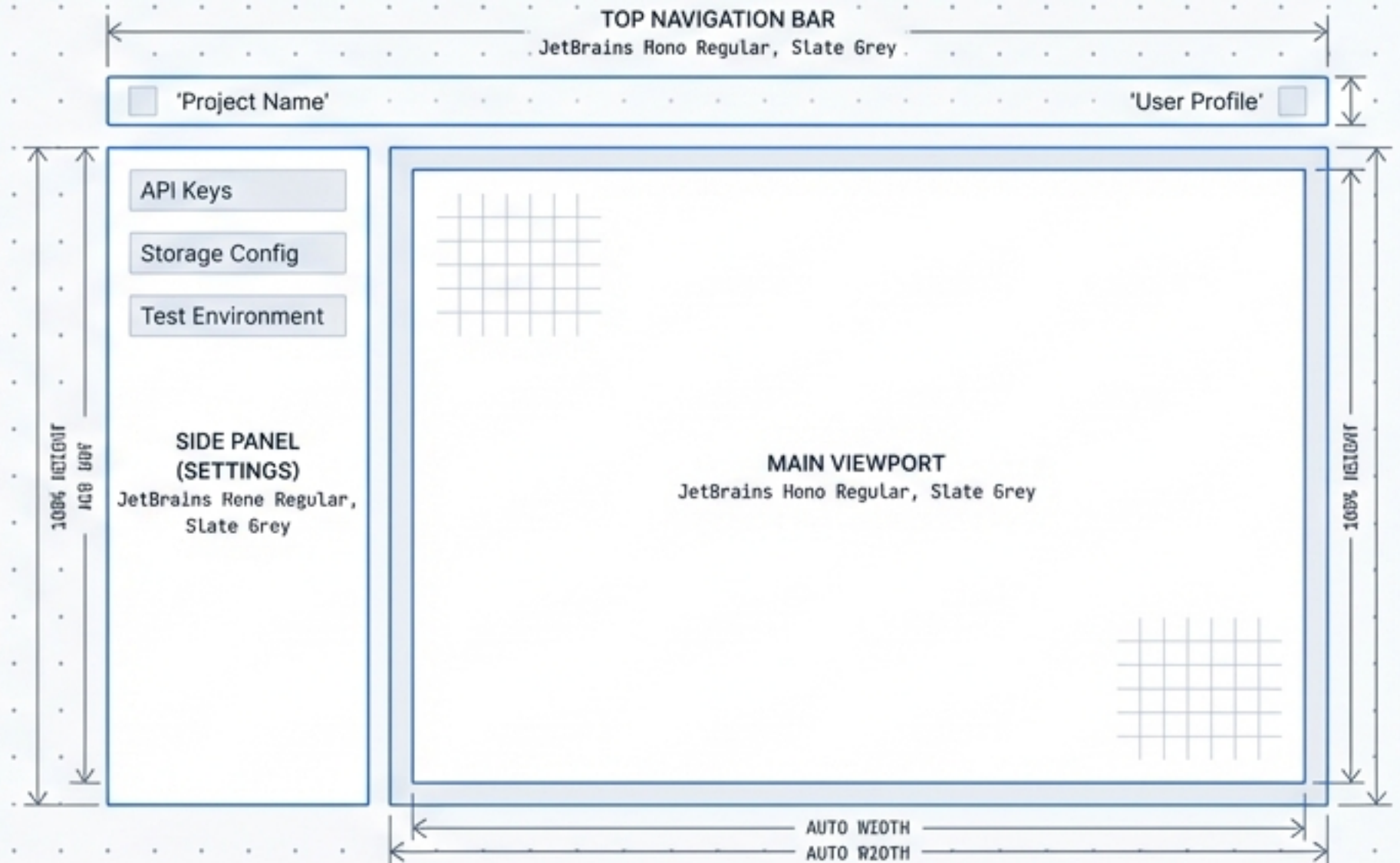
Configure API Keys



Persistent Local Storage



Load/Save Round-trip



[STATUS: COMPLETE]

Roadmap Phases 2-4 — The Builder's Toolkit

Phase 2 (Browsing & Profiles)

Site Browser list view, Profile Manager creation/editing, MitmProxy lookup

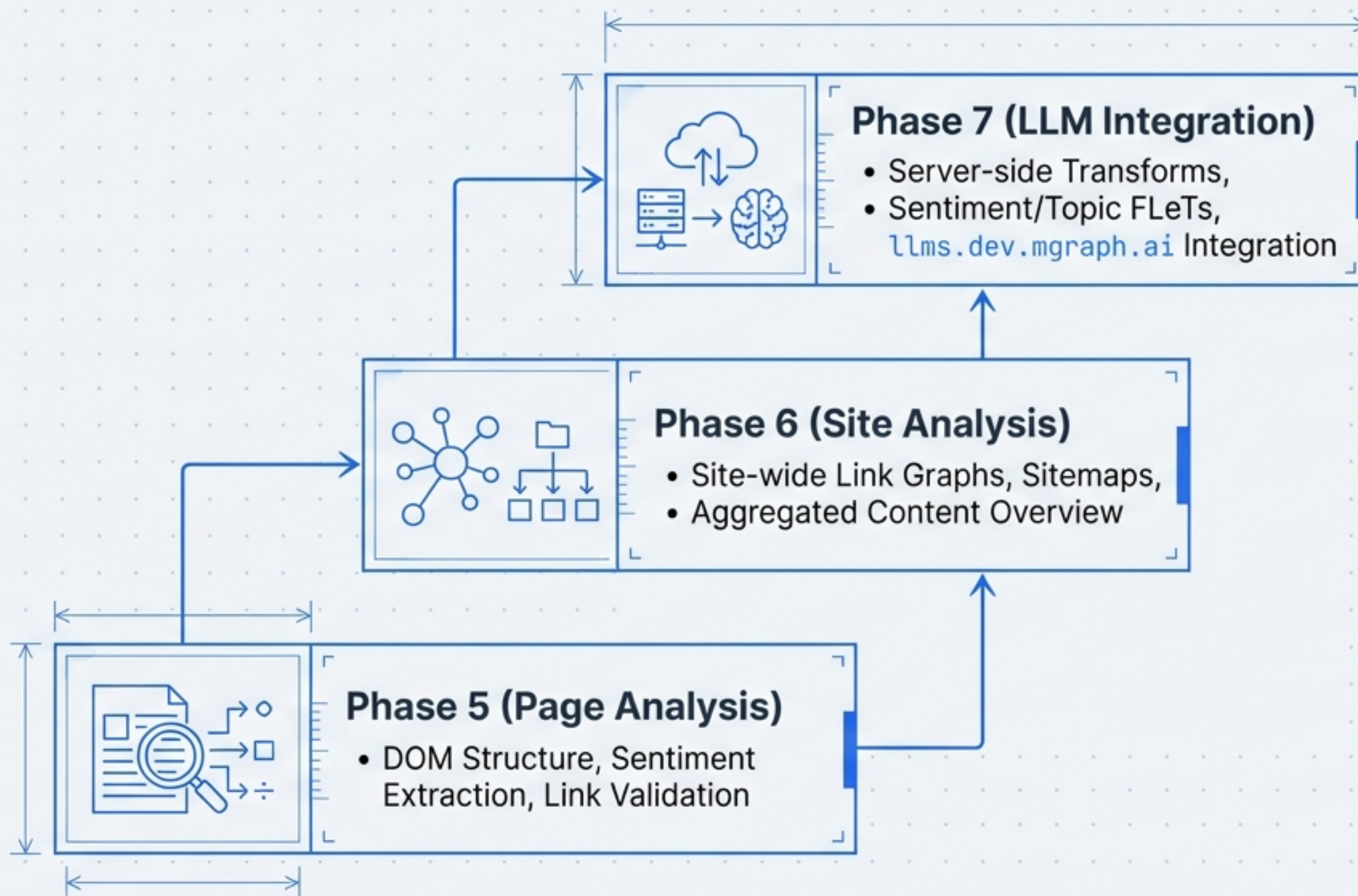
Phase 3 (The Workbench)

Split-pane Before/After interface, Real-time profile application, Batch logic

Phase 4 (API Explorer)

Request builder, Response viewer, History logging

Roadmap Phases 5-7 — Intelligence & Analysis



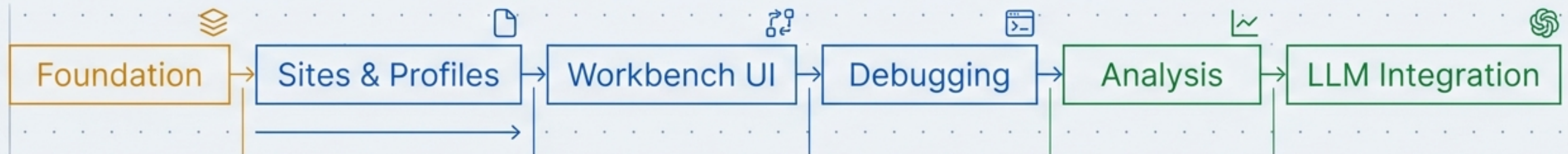
STATUS: PLANNED ✓

Execution Strategy & Next Steps

Iterative Flow Development (IFD)

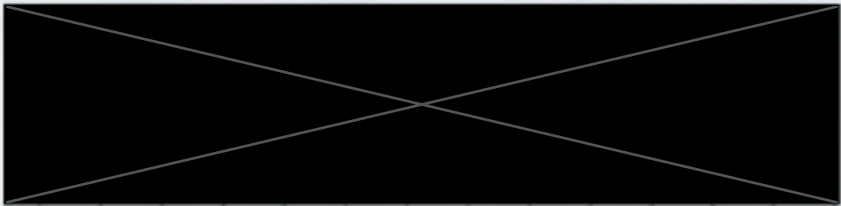
The Critical Path

1. **Immediate:** Start Phase 1 (Shell, Settings, Basic Load/Save) [PH1_START]
[SHELL_INIT]
2. **Validate:** Prove API integration with HTML Graph [API_CHECK: HTML_GRAPH]
[INTEGRATION_TEST] [INTEGRATION_TEST]
3. **Expand:** Move to Site Browser and Profile Manager [PH2_MIGRATION]
[BROWSER_UI]



⚙️ Each mini-app maintains its own version tracking to ensure modular progress.

Appendix: API Specifications

	Profile Management	
 	<code>GET /flet-html-domain/profiles/{ns}/{profile_id}</code> <code>POST .../apply/site/{domain}</code>	
	Site Browsing	
	<code>GET /cache-entity/{ns}/entities/site/{domain}</code>	
	Analysis Endpoints	
	<code>POST .../html/analyze/{ns}/content</code>	[LLM Analysis] 
	<code>POST .../site/graph/{ns}/links/{domain}</code>	[Site Link Graph] 
	<code>POST .../html/graph/{ns}/dom-tree</code>	[Generate DOM Graph] 